

A comparison of FFT-based techniques for Doppler velocity estimation in SuperDARN radars

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Abstract—In radar applications, the target velocity is commonly determined using the Doppler effect. By comparing the transmit-receive differential frequency, the Doppler frequency shift can be measured, and as a result, the target velocity can be determined.

The Tasman International Geospace Environment Radars (TIGER) form part of an international network of similar HF radars called Super Dual Auroral Radar Network (SuperDARN) which explores the impact of solar disturbances on the Earth's upper atmosphere. These radars utilise an Auto Correlation Function (ACF) to measure the changing phase of the ACF between lag times to determine the Doppler frequency and the target velocity. Measured velocity results can show large, and sometimes unrealistic errors.

As part of the development of the third TIGER radar at Buckland Park, Adelaide, South Australia, a Spectrum Difference Function (SDF) technique for measuring velocity has been proposed as a means for cross-checking results. The SDF technique uses the Fast Fourier Transform (FFT) to calculate the transmit and receive signal magnitude spectrums which are then compared to find the Doppler frequency. In this paper the developed technique is compared to existing interpolation techniques using SuperDARN radar parameters. Simulation results show that the accuracy and computational complexity of the SDF technique are comparable to those of other techniques using FFT.

I. INTRODUCTION

The Tasman International Geospace Environment Radars (TIGER) form part of an international network of similar HF radars called Super Dual Auroral Radar Network (SuperDARN). This network explores the impact of solar disturbances on Earth by monitoring the location of aurora and related phenomena occurring in the ionosphere [1]. The ACF (Auto Correlation Function) technique is utilised by SuperDARN to measure: velocity, spectral width and signal power [2], [3]. In this technique, the measured phases of the complex ACF in all good lags are calculated and a phase fitting procedure is then implemented to generate true phases. The velocity and the velocity error are estimated based on the changing phase between lags [2], [3].

The current ACF technique, however, occasionally produces unrealistic velocity estimates [4]. An independent calculation is therefore desirable in evaluating the velocity estimation from the ACF technique. With the actual velocity of the target unknown, the reason for the unrealistic velocity estimates can be evaluated using either of two methods: (i) using a realistic model to evaluate the error, and (ii) cross-checking by another technique whose validity has been confirmed.

Following the former approach, there have been several attempts to develop realistic simulators, which are used to evaluate the accuracy of the ACF technique. The most recent one, using a scatter model [5], has shown that the FITACF routine introduces a large error at high velocities [6]. However, it raises the question of the model validity which is based on a number of assumptions and difficult to prove.

The latter approach is one of the focuses in the development of the third TIGER radar. The new technique for velocity estimation should be applicable for all TIGER and SuperDARN radars. For cross-checking, it would be deployed in parallel with the current ACF technique in the radar system and therefore should not consume significant hardware and processing time. The new technique is expected to have higher or at least an equal level of accuracy compared to the existing one.

With the rapid development and availability of high end FPGA, DSP and reconfigurable devices, FFT-based techniques for Doppler Frequency estimation are in favor. The recent trend is the utilisation of the Fast Fourier Transform (FFT) to produce the received (Rx) signal spectrum, from which the peak is located and therefore Rx frequency. With a given Tx frequency, that means Doppler frequency and target velocity can be deduced. However, the real peak of spectrum in most instances lies between two FFT samples. The bigger the gap between these two samples, hereafter known as frequency bin width of the FFT, the larger the error in the frequency estimation. In order to achieve higher estimation accuracy, the common solutions involve collecting more data samples or zero-padding of the received signal, thereby reducing the bin width between FFT samples. The drawback of these solutions is that the computation complexity is increased. A more efficient method is the interpolation approach [7]. This approach utilises several FFT samples near the peak for interpolating the carrier frequency and consists of two steps. In the first, the maximum value of the FFT samples is located. This value is the coarse estimation, which is the starting point for the subsequent fine estimation. In the second step, several FFT samples around the maximum sample are taken for interpolating the real peak frequency. In [8], Provencher used only two samples for interpolation, but three samples are more frequently used as suggested in [9], [10], [11] and [12]. In [13] and more recently in [14] it was shown that the interpolation algorithm, hereafter known as the Spectrum

Peak Location (SPL) technique, using three samples, closely follows the Cramer-Rao lower bound in high SNR regions. Moreover, with the use of only three FFT samples, the SPL technique is very computationally efficient.

The SPL technique, however, has several limitations. Firstly, in order to achieve high accuracy, it requires all three samples to lie within the main spectral beam, which is not always the case. Secondly, using only three samples at a small region in the spectrum potentially runs the risk of inaccurate frequency estimation in strong noise and interference environments. In such situations, more samples in a larger region of the spectrum could improve the accuracy. Thirdly, in [7], Lyons summarized SPL techniques using 3 samples, rectangular and other windowing of signals for both FFT magnitude and complex samples. Lyons then concluded that these algorithms can achieve a high level of accuracy and are applicable when a majority of the signal's energy lies within a single FFT bin and the FFT spectral samples are not substantially contaminated by leakage from other spectral components. In conditions where the spectrum is broad, the accuracy of the SPL technique reduces rapidly. Furthermore, the SPL technique does not utilise the Tx spectrum. While this is probably not a limitation of the SPL technique, clearly there is a correlation between the Tx and Rx spectrum which can be exploited to provide additional information about Doppler frequency.

As part of the development of the third TIGER radar at Buckland Park [15], a Spectrum Difference Function (SDF) technique for measuring velocity has been proposed as a means of cross-checking results [16]. In the proposed technique, the correlation between Tx and Rx spectrums is taken into account to find the Doppler frequency and more FFT samples are used to reduce the influence of noise. More specifically, the transmit signal magnitude spectrum is considered as a reference for direct comparison with the received signal magnitude spectrum. The Spectrum Difference Function (SDF) is derived from this differential. Mathematical proof shows that the gradient of the SDF in the vicinity of the carrier frequency is proportional to the target Doppler frequency [16]. Therefore, by calculating the gradient of the SDF, the target Doppler frequency and consequently target velocity can be deduced. Several technical aspects of the SDF technique, such as improving the accuracy and reducing the computation complexity, have been discussed in [17] and [18].

The aim of this paper is to compare the developed SDF technique with the SPL technique for the application in SuperDARN radar platforms. The remainder of this paper is organized as follows: Section II briefly reviews the SPL and SDF techniques. Sections III compares MATLAB simulation results of the SDF and SPL techniques affected by Gaussian noise for a range of conditions, followed by a discussion. The conclusion is provided in Section IV.

II. REVIEW OF SPECTRUM PEAK LOCATION AND SPECTRUM DIFFERENCE FUNCTION TECHNIQUES

A. Spectrum Peak Location technique

Assuming we have N spectral samples from a N -point FFT of the Rx signal. The SPL technique firstly locates the peak in the FFT output magnitude samples $|X(k)|$, the index k of the peak gives a coarse estimation of the Rx frequency. Then a parabolic (second-order polynomial) fit is implemented over the peak and the two adjacent FFT samples (one for each side). This process calculates the correction factor Δk for fine estimation. The mathematical proof of the SPL technique in [19] shows that:

$$\Delta k = \frac{|X(k+1)| - |X(k-1)|}{4|X(k)| - 2|X(k+1)| - 2|X(k-1)|} \quad (1)$$

The frequency of the Rx signal can therefore be estimated as:

$$f_{peak} = (k + \Delta k) \frac{f_s}{N} \quad (2)$$

A modified version of the SPL technique using complex FFT samples was introduced by [12] and shows that:

$$\Delta k = \text{Re}\left(\frac{X(k-1) - X(k+1)}{2X(k) - X(k+1) - X(k-1)}\right) \quad (3)$$

There are other modified equations published to calculate the correction factor, depending on the window function applied to the Rx signal [7], [12]. In this paper, a rectangular window is applied, therefore only equations 1 and 3 are tested in the SPL technique. The first, which uses magnitude spectrum samples in equation (1), is named SPL1. The second uses complex spectrum samples in equation (3), is named SPL2.

The process of determining Doppler frequency f_d using the SPL technique therefore consists of the following steps:

- 1) Calculate the spectrum of the Rx pulse by FFT
- 2) Scan the spectrum to locate the peak
- 3) Use equation (1) or (3) to find correction factor and then equation (2) to calculate the Rx frequency
- 4) Subtract the Tx from the Rx frequency to find f_d

The computational task complexity of the SPL technique is therefore almost exclusively occupied by the FFT.

B. Spectrum Difference Function technique

The SDF technique was introduced in [16]. The Tx and Rx signal magnitude spectrums are determined by calculating their respective Fourier transforms, and their difference is the SDF. It is shown in [16] that the gradient ($Grad$) of the SDF, in the neighbourhood of f_c , is linearly proportional to the target Doppler Frequency f_d :

$$Grad = \frac{\pi^2 T^3}{6} f_d \quad (4)$$

where T is the pulse width of the radar signal.

From equation (4), if the value of $Grad$ is calculated, the Doppler frequency, and hence the target velocity, can be determined. In [16], the process of determining f_d using the SDF technique consists of the following steps:

Step 1: Calculate the spectrum of the Tx signal (this is the reference spectrum and is calculated only a single time)

Step 2: For one Rx pulse (reflected signal for a single range gate)

- 1) Calculate the spectrum of the Rx pulse by FFT
- 2) Standardise the Rx signal. As discussed in [16], the SDF technique requires that the magnitude of the Tx and Rx signals be equal. The standardisation process is implemented in the frequency domain
- 3) Calculate the SDF
- 4) Use the Least Mean Square method and linear fit to calculate Gradient of the SDF in the vicinity of carrier frequency
- 5) Calculate f_d using equation (4)

The SDF technique therefore increases the computation complexity in comparison to the SPL technique. However, in applications which necessitate a large number of FFT samples, the computation required by the two techniques is comparable. This is because a majority of the computational task is the FFT calculation.

III. SIMULATION RESULTS

A. Results

In this section, the TIGER radar is utilised as a test platform and all of the radar parameters are consistent with standard operational parameters.

The transmit signal is simulated in MATLAB with the following parameters: pulse length $T = 300 \mu\text{s}$ and $f_c = 8 \text{ MHz}$. With a sampling frequency of $f_s = 125 \text{ MHz}$, a single pulse contains 37,500 samples. A single period is 100 mS and thus contains 12.5 million samples. The Rx signal is simulated by adding Gaussian noise and introducing a random generated Doppler frequency, phase and magnitude to the Tx signal.

If the FFT of the Rx pulse is taken, the number of samples in the FFT is $N = 37,500$, the frequency bin width achieved is therefore

$$f_w = \frac{f_s}{N} = 3333 \text{ Hz}$$

The SDF technique is not applicable in this arrangement because it utilises the linear region of the SDF in the vicinity of the carrier frequency. The linear region has been shown to be approximately 4000 Hz [16]. Therefore, this region covers only two samples of the FFT and is an insufficient number for the SDF technique to achieve high level of accuracy. The SPL technique, where the error is proportional to the frequency bin width of the FFT, results in an error in the order of tens of Hz. This level of frequency accuracy is inadequate for the TIGER radar application.

The zero-padding is thus necessary in both techniques to achieve higher accuracy. When an Rx pulse is processed, the entire 100 mS pulse sequence (1 period) is zero padded. With a sampling frequency of $f_s = 125 \text{ MHz}$ and the length of the FFT is $N = 12,500,000$, the frequency bin width achieved is

$$f_w = \frac{f_s}{N} = 10 \text{ Hz}$$

The two techniques are tested with the same random Rx signal generated in MATLAB. For the SPL technique, the magnitude spectrum (SPL1) and complex spectrum (SPL2) versions are tested. Ten sets of 10,000 random Rx signals are generated for the simulation. For a single set of 10,000 random Rx signals, two variables are produced; the mean value of error and the root mean square of the error (RMSE). These are both taken as the quality benchmark for the remainder of this paper.

Simulation results when the $SNR = 20 \text{ dB}$ are shown in Figure 1 and Figure 2.

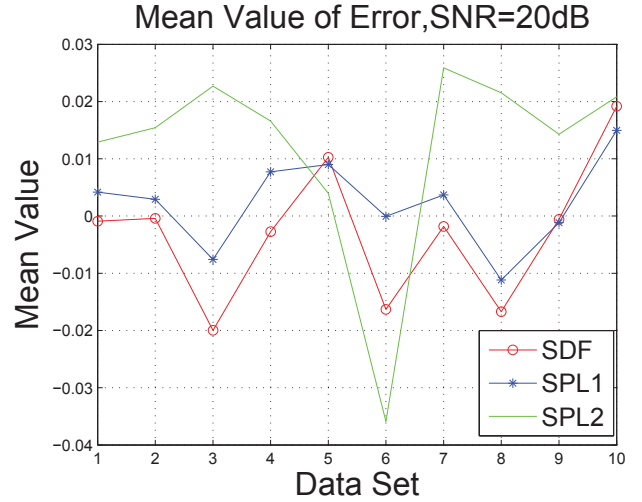


Fig. 1. Mean Value of Error

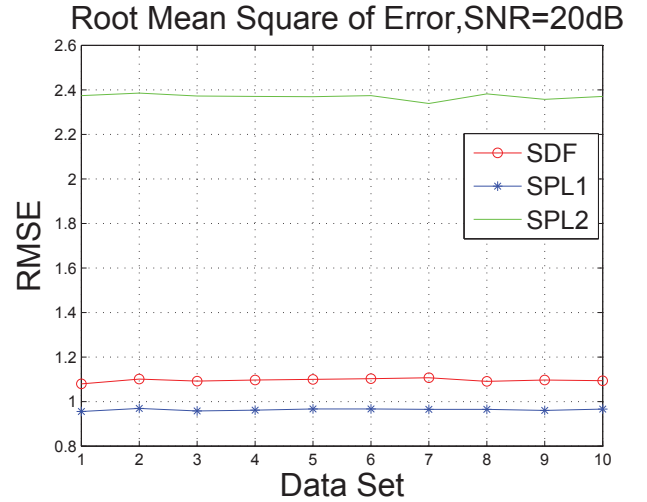


Fig. 2. Root Mean Square of Error

In Figure 1, all three techniques show that the mean value of the error fluctuates closely around zero, showing unbiased estimation. In Figure 2, the SDF and SPL1 are superior compared to SPL2 in terms of RMSE. If the SDF or SPL1 is applied where the $SNR=20 \text{ dB}$, an estimated accuracy of approximately 1 Hz can be achieved.

Simulation results when $SNR = 10dB$ are shown in Figures 3, 4 and for a $SNR = 0dB$ in Figures 5, 6.

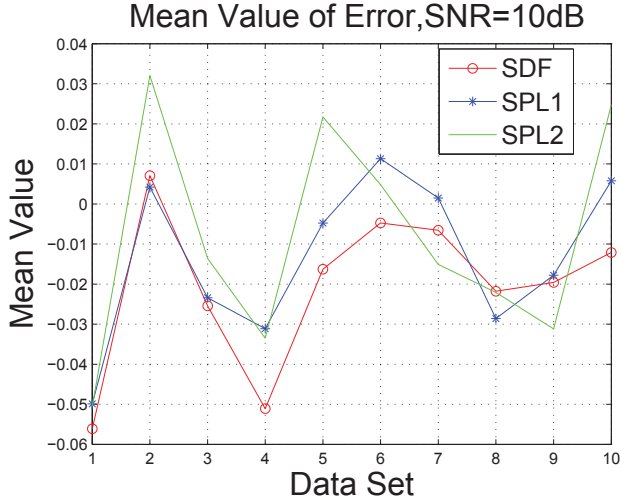


Fig. 3. Mean Value of Error

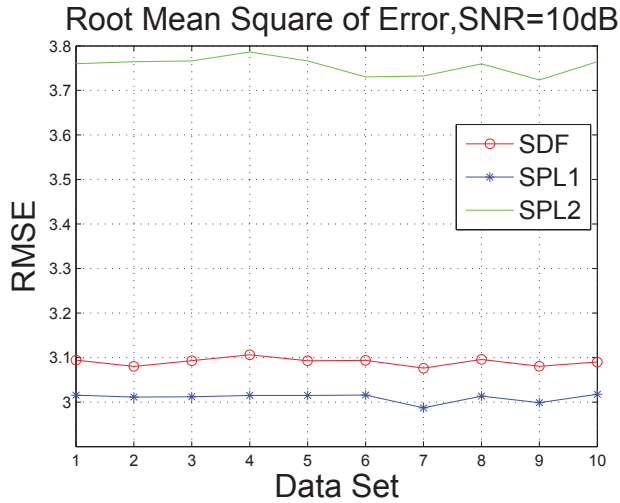


Fig. 4. Root Mean Square of Error

The results in Figures 1 to 6 show that the SPL1 and SDF consistently outperform the SPL2 technique. Furthermore, the SPL1 average deviation is marginally smaller than that of the SDF technique under these conditions.

B. Discussion

The existing velocity resolvment technique utilised by SuperDARN radars, ACF, has no means of cross checking Doppler velocities and has been shown to produce large errors at times. For this reason additional methods of determining velocities have been investigated. The results show that the newly developed SDF technique achieves comparable results in both accuracy and computational requirements as the SPL technique. As both techniques are comparable, they will both

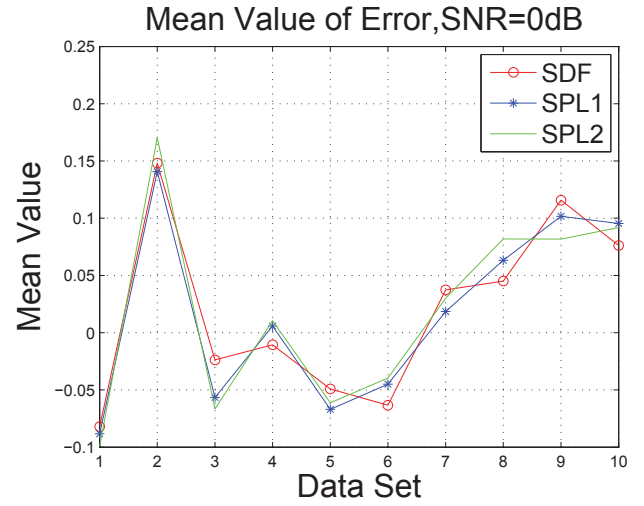


Fig. 5. Mean Value of Error

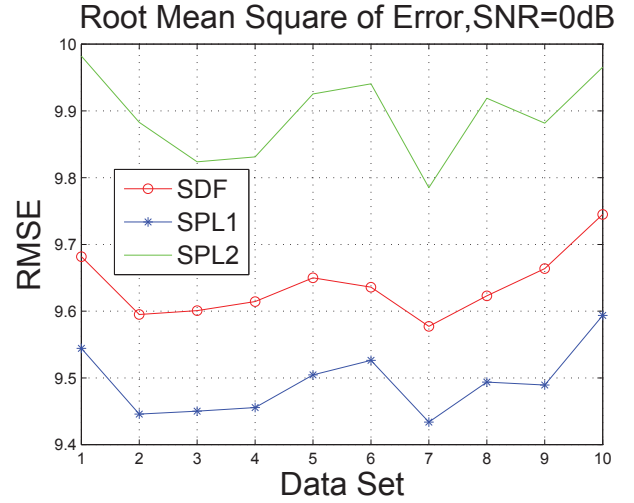


Fig. 6. Root Mean Square of Error

need to be assessed for the application of Doppler velocity resolvment on SuperDARN radar platforms. Since the TIGER-3 radar is the testing platform of the techniques presented in this paper, implementation directly at the RF frequency with zero padding, requires an FFT of 12.5×10^6 samples. This is a significantly large computational requirement and is not currently feasible in the existing digital TIGER radar configuration. However, as discussed in [18], the computational length of the FFT can be significantly reduced by moving the processing from the RF frequency band to a low IF band, followed by decimation. Thus, the computational length of the FFT can be reduced significantly to the order of 10,000 samples without compromising the validity of the technique [18]. An FFT of 10,000 samples can be implemented for this application with ease in all high end Field Programmable Gate Arrays (FPGA) devices.

At this level of the FFT length, the computational complexity of the SDF and SPL can be considered comparable. Yet, the SDF technique requires additional computation for the

standardization process, SDF calculation and fitting procedure. However, the additional computation can be considered negligible in comparison to that required by the FFT.

All of the simulations in this paper have been implemented with a single frequency component Rx signal using the TIGER radar platform. In this scenario, both the SDF and the SPL1 achieve high levels of accuracy. However, the Rx signal more commonly consists of multiple frequency components and has broad spectral width. Further research needs to be conducted to evaluate both techniques under such parameters.

IV. CONCLUSION

This paper proposes the use of two Doppler velocity estimation techniques, namely SDF and SPL, for cross checking the current ACF technique in SuperDARN radars. The SDF technique developed by the authors is compared against an existing interpolation technique for the resolution of Doppler velocities. Both techniques utilise the FFT to estimate the Doppler frequency and are compared using SuperDARN radar operational parameters. Simulation results show comparable performance in both accuracy and computational complexity for the SDF and SPL techniques using the magnitude spectrum. The two techniques are essential for the study and cross checking of velocity measurements using SuperDARN radars. At present, these can be implemented in high end FPGA devices resulting in high levels of accuracy. Further investigation of both techniques using signals with spectrum width commonly seen by SuperDARN radars is therefore necessary to ascertain their superiority over the ACF.

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